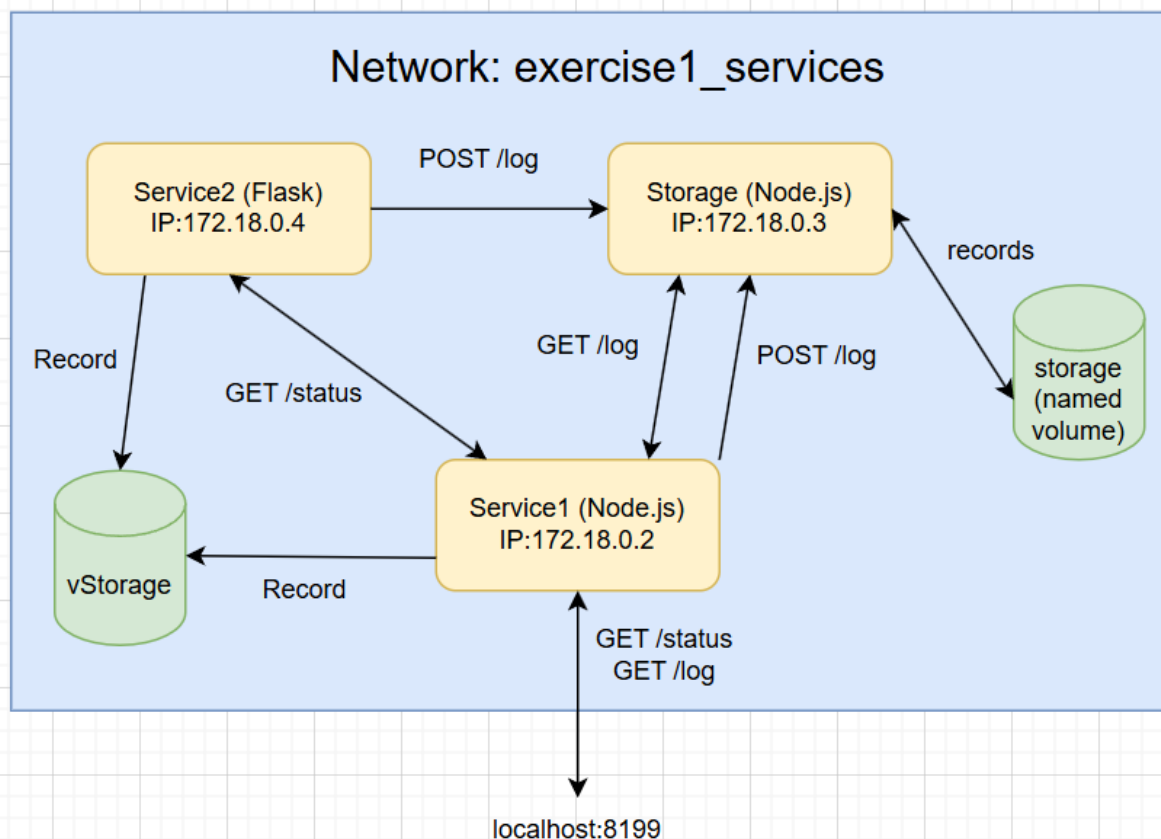


Report – Exercise 1

Platform Information

- **Hardware / Virtual Machine:** HP Laptop 14s-fq0xxx / Tested with Oracle VirtualBox (Ubuntu 24.04.3 LTS)
- **Operating System:** Microsoft Windows 10 Home 22H2
- **Docker version:** Docker version 27.3.1, build ce12230
- **Docker Compose version:** Docker Compose version v2.29.7-desktop.1

System Architecture Diagram



Picture 1. Architecture of the system

Analysis of Status Records

The records produced by Service1 and Service2 measure how long the respective service container has been running and how much free space is available on the container's internal filesystem. These data are relevant if the goal is to monitor only service-specific uptime and disk usage, but otherwise they are limited because they don't provide information about the host's free disk space or the status of the Storage service.

As an improvement, information about the host's free disk space should also be collected, so that the storage used by volumes can be monitored more accurately. Additionally, data on the Storage service's status should be gathered to enable more accurate monitoring of the entire system.

Persistent Storage Solutions

Simple persistent storage (vStorage):

- **Advantages:** Inspecting and clearing stored data is simple, data persists across container restarts
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- **Disadvantage:** Writes directly to the host file, which introduces security risks and scalability issues. The service depends on the host's filesystem (moving the directory can cause problems), race conditions can occur if two different services write to the same file simultaneously

Separate container storage (named volume):

- **Advantages:** Docker fully manages the volume, responsibility is isolated to a single service, safer than writing directly to the host file, data persists across sessions
- **Disadvantage:** Inspecting the contents of the volume is more difficult and requires more knowledge of Docker and its management

Cleaning Persistent Storage

Cleaning the simple storage (VStorage):

Clean up the persistent volume file manually from `./vStorage\records.txt` or use command `> ./vStorage/records.txt`

Cleaning the separate container storage (named volume):

Delete the entire volume (when volume is not in use): `docker volume rm exercise1_storage`

Clear only the content of the storage file while the storage container is running:

`docker exec storage sh -c ' > /storage/records.txt'`

Cleaning up both persistent storages with a single command:

`> ./vStorage/records.txt && docker exec storage sh -c ' > /storage/records.txt'`

Challenges and main problems

The most challenging part was coding the functionality of Service2 because using Flask was new to me and I hadn't used Python in a long time. Working with volumes also required a bit of learning to refresh my memory. Other than that I didn't face any significant challenges or problems.